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GNIOT Institute of Management Studies – PGDM (Batch 2022-24)
(TRIMESTER – I) END TERM EXAMINATION, SEPTEMBER - 2022
STATISTICS FOR BUSINESS DECISION

[Time: 2 Hours 30 Minutes]

[Maximum Marks: 100]

INSTRUCTIONS:

- (i) There are **THREE** sections in this question paper.
- (ii) Attempt questions from each section as per the directions.
- (iii) Make suitable assumptions wherever necessary.

SECTION-A**(Short Answer Type Questions, do not exceed 100 words)****Q.1. Attempt all parts of the following:****(10×3=30)**

- (a) Explain census.
- (b) Distinguish between primary and secondary data.
- (c) Differentiate between bar chart and histogram.
- (d) Illustrate when median is preferred over mean.
- (e) Describe levels of data measurement.
- (f) Identify the methods for measuring central tendency.
- (g) State the range of skewness and interpret it.
- (h) Interpret lower and higher standard deviation. Which is better?
- (i) Distinguish between z test and t test.
- (j) Determine the probability of getting 2 heads in tossing two coins.

SECTION-B**(Long Answer Type Questions)****Attempt all questions of the following:****(4×10=40)****Q.2. (a)** The following are the starting monthly salaries of the 11 students of your class on joining HUL:

Salaries in Rs '000										
33	32	30	28	29	27	25	34	26	35	36

On completion of one year, HUL gave annual bonus of Rs 60,000 to all those whose monthly salary was less than Rs 30,000 and Rs 1,20,000 to those having monthly salary equal to or higher than Rs 30,000. Determine the monthly Median of salary and bonus paid.

OR (Both b and c)(b) Measure Skewness and Kurtosis if $\mu_2 = 0.918$, $\mu_3 = 0$ and $\mu_4 = 2.533$. Interpret your results.

(c) In a class of 50 students, 10 have failed and their average marks are 25. The total marks scored by the entire class are 2810. Find the average of those who have passed.

Q.3. (a) A brokerage survey reports that 50% of individual investors have used a discount broker, that is, one which does not charge full commission. In a random sample of 9 individuals, what is the probability that exactly 5 of the sampled individuals have used a discount broker?

OR

(b) Determine the 77th percentile of the following eight numbers: 100, 122, 167, 123, 125, 99, 87, 88.

Q.4. (a) Test significance between Commitment to Work and Happiness from the data given below:

Subject	Commitment to work	Happiness
1	20	9
2	15	10
3	10	10
4	10	9
5	15	8
6	8	7
7	8	7
8	7	8
9	11	9
10	11	10

Values of “r” are given on the last page.

OR

(b) Lakme employed you as a Sales Manager for NCR region. You have been asked to predict sales for the year 2020 based on the data of the previous years. Develop a regression model based on 25 salespersons reporting to you in the year 2020. Also predict sales for the year 2020.

Furthermore, on doing calculations, you obtained the following values:

Total of Sales (in last 10 years) = 1236

Total of Number of Salespersons employed = 172

Total of (Sales x Number of Salespersons) = 21,358

Total of (Sales)² = 1,53,928

Total of (Number of Salespersons employed)² = 3104

(All sales figures are in Rupees million)

Sales (in last 10 years)	120	125	118	115	100	130	140	135	130	123
Number of salespersons employed	10	15	12	18	20	21	22	20	15	19

Q.5. (a) A distribution of numbers is approximately bell shaped. If the mean of the numbers is 125 and the standard deviation is 12,

- Between what two numbers would approximately 68% of the values fall?
- What percentage of values would lie beyond 161?
- Between what two numbers would 95% of the values fall?
- What percentage of values would lie below 113?

OR

(b) Of 1000 assembled components, 10 have a working defect and 20 have a structural defect. There is a good reason to assume that no component has both the defects. What is the probability that a randomly chosen component will have either type of defect?

SECTION-C

Q.6. Attempt all parts of the following:

(2×15=30)

6(a) A random sample of people in your colony was taken by you and is classified by age group as follows:

Age group	No. of people surveyed
10 - 15	6
15 - 20	22
20 - 25	35
25 - 30	29
30 - 35	16
35 - 40	8
40 - 45	4
45 - 50	2

Determine Mean, Median, and Mean Deviation.

6(b) Using the above data, also calculate Standard Deviation and Variance. Give interpretation of your results obtained in Q6(a) and Q6(b).

Values of r for the .05 and .01 Levels of Significance

$df(N - 2)$	.05	.01	$df(N - 2)$	.05	.01
1	.997	1.000	31	.344	.442
2	.950	.990	32	.339	.436
3	.878	.959	33	.334	.430
4	.812	.917	34	.329	.424
5	.755	.875	35	.325	.418
6	.707	.834	36	.320	.413
7	.666	.798	37	.316	.408
8	.632	.765	38	.312	.403
9	.602	.735	39	.308	.398
10	.576	.708	40	.304	.393
11	.553	.684	41	.301	.389
12	.533	.661	42	.297	.384
13	.514	.641	43	.294	.380
14	.497	.623	44	.291	.376
15	.482	.606	45	.288	.372
16	.468	.590	46	.285	.368
17	.456	.575	47	.282	.365
18	.444	.562	48	.279	.361
19	.433	.549	49	.276	.358
20	.423	.537	50	.273	.354
21	.413	.526	60	.250	.325
22	.404	.515	70	.232	.302
23	.396	.505	80	.217	.283
24	.388	.496	90	.205	.267
25	.381	.487	100	.195	.254
26	.374	.479	200	.138	.181
27	.367	.471	300	.113	.148
28	.361	.463	400	.098	.128
29	.355	.456	500	.088	.115
30	.349	.449	1000	.062	.081